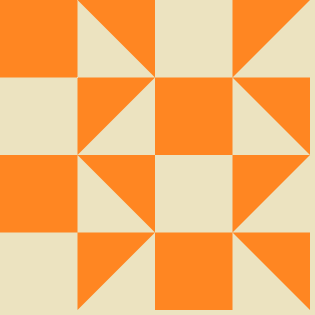


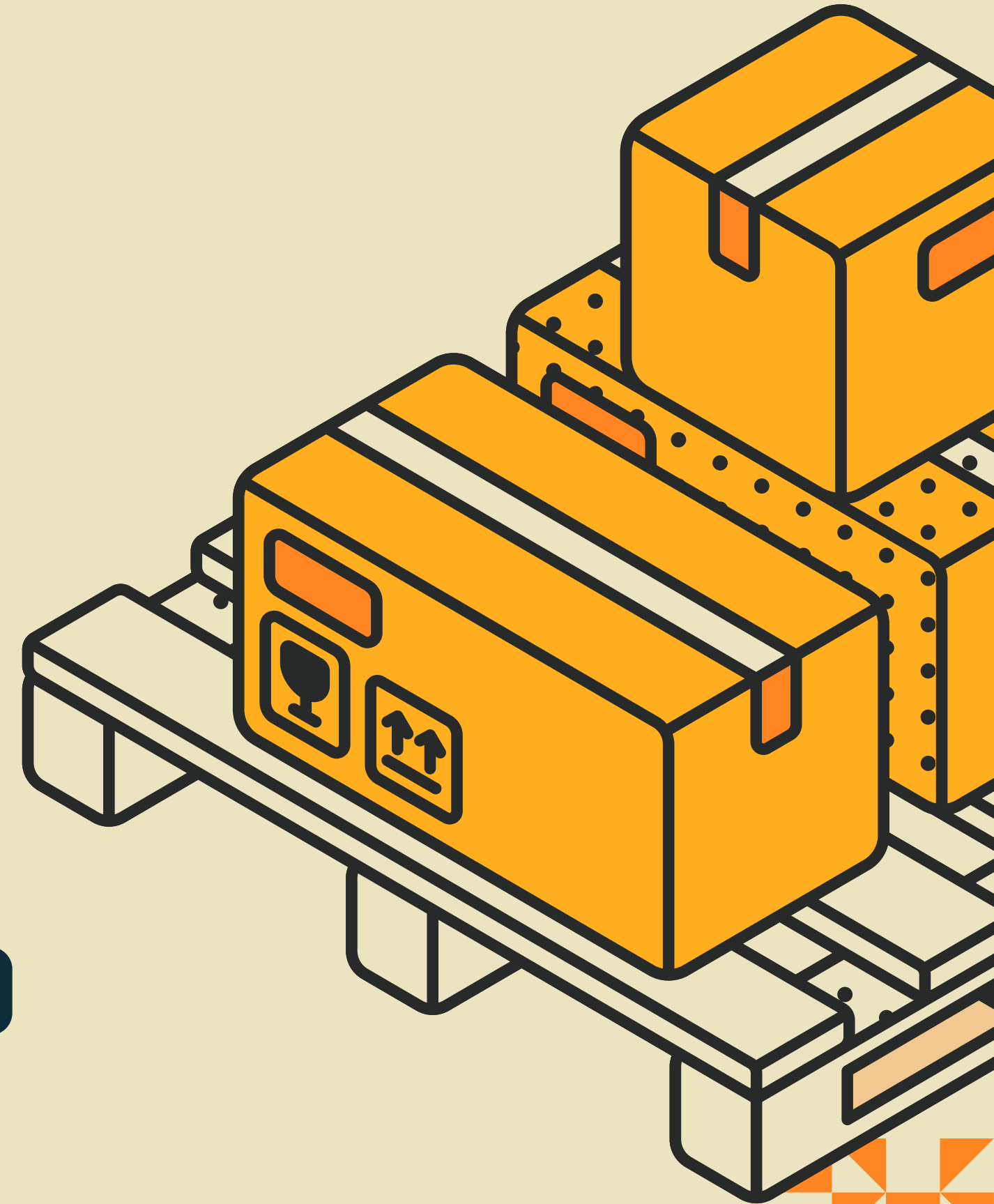


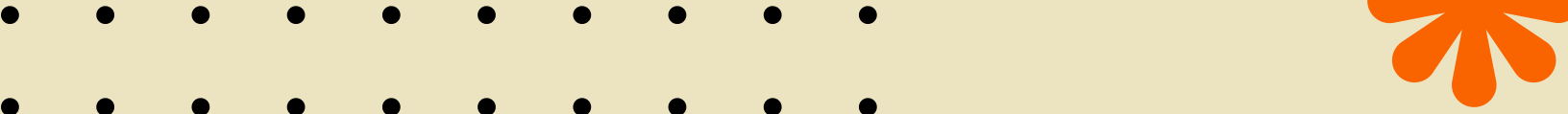
AUGUST 2026 | EXECUTIVE PERFORMANCE REPORT

UNIVERSAL SUPPLY CHAIN & FORECAST ANALYTICS

END-TO-END SQL DATA WAREHOUSE & BI AUDIT
(PHARMA, FMCG & E-COMMERCE)

PREPARED BY: TRIVENI SHARMA | SUPPLY CHAIN DATA ANALYST

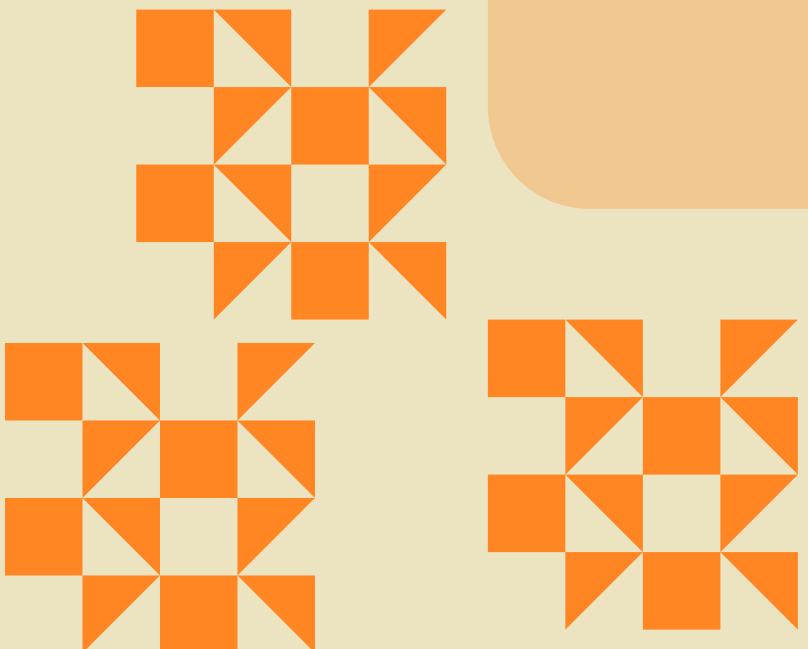


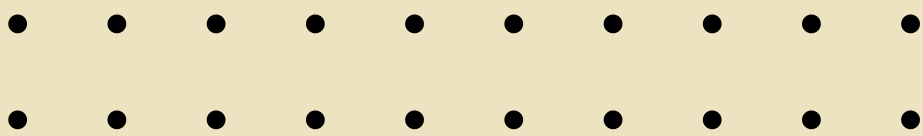


COMPANY OVERVIEW & OPERATING SCOPE



Universal Supply Chain Network is a multi-category global enterprise operating across four key geographical regions: APAC (India), EU (Switzerland), NA (USA), and LATAM (Brazil). The company manages a complex logistics ecosystem spanning high-value Pharma Cold-Chain (mRNA Vaccines & Insulin), FMCG Fresh (Dairy & Cheese), Electronics, and Personal Care.





DATA WAREHOUSE ARCHITECTURE



dim_products



dim_suppliers



dim_warehouses



fact_customer_shipments



fact_purchase_orders



fact_forecast_monthly

+ 4 SUPPORTING FINANCIAL & DATE TABLES: DIM_DATE,
FACT_GROSS_PRICE, FACT_MANUFACTURING_COST,
FACT_FREIGHT_COST



STRATEGIC MISSION

To transform our global supply chain network across Pharma Cold-Chain, FMCG Fresh, and Electronics into a **resilient, data-driven operation**—achieving zero inventory spoilage, eliminating customer delivery delays, and maximizing supplier OTIF reliability.

KEY PERFORMANCE GOALS

1. ❄️ **Cold-Chain Quality Assurance:** Eliminate transit temperature excursions on high-value bio-pharma SKUs (-25°C to 15°C bounds).
2. 🚚 **Fulfillment SLA Optimization:** Eradicate customer delivery delays across global fulfillment hubs to avoid late penalty costs.
3. 📦 **Supplier OTIF Compliance:** Audit global vendors against an 80% On-Time In-Full (OTIF) delivery threshold to prevent dark store stockouts.
4. 📈 **Demand Forecast Accuracy:** Eliminate error cancellation variances (using MAPE / Absolute Error) to optimize safety stock levels.



Q1) COLD-CHAIN PRODUCT CATALOG AUDIT

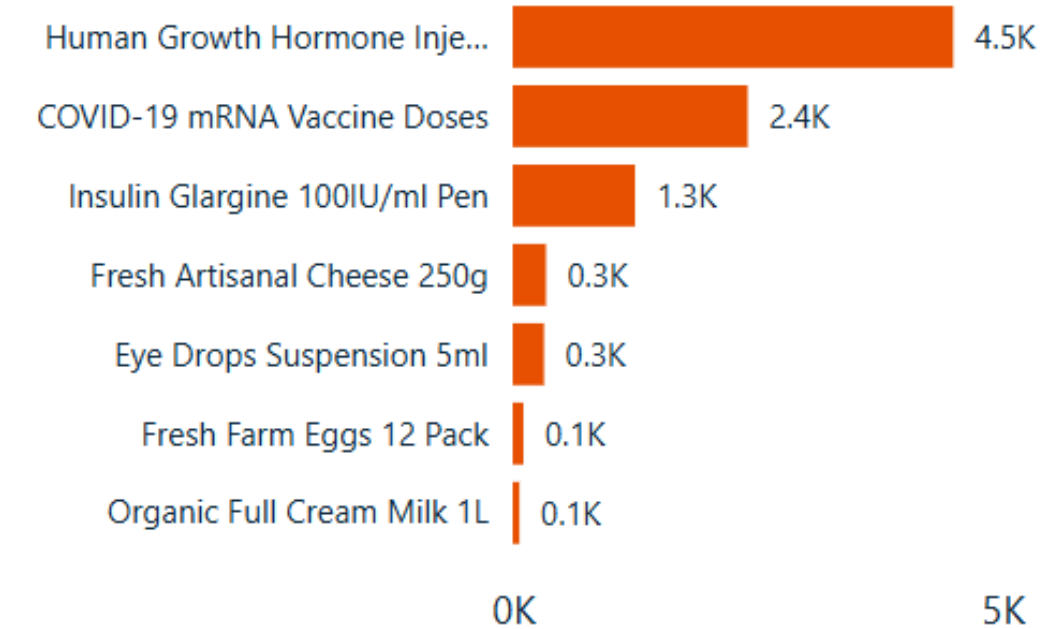
“Our pharmacy and fresh grocery category managers need a catalog report of all products that require temperature-controlled storage. Write a query to display a combined product detail string , its unit cost, and its minimum and maximum safe storage temperatures in Celsius. Filter the results to include only temperature-sensitive items, and order the list from the highest unit cost to the lowest.”

```
SELECT
product_id,
concat_WS('|', product_name,category) AS Product_profile,
unit_cost,
target_temp_min_c,
target_temp_max_c
FROM dim_products
WHERE (Category LIKE '%PHARMA%' OR category LIKE '%FRESH%')
AND is_cold_chain = 1
ORDER BY unit_cost DESC;
```

Result

Identified 7 high-value cold-chain products across Pharma and FMCG
Fresh categories with specialized temperature control requirements
ranging from -25°C deep freeze (mRNA Vaccines) to 15°C ambient cold.

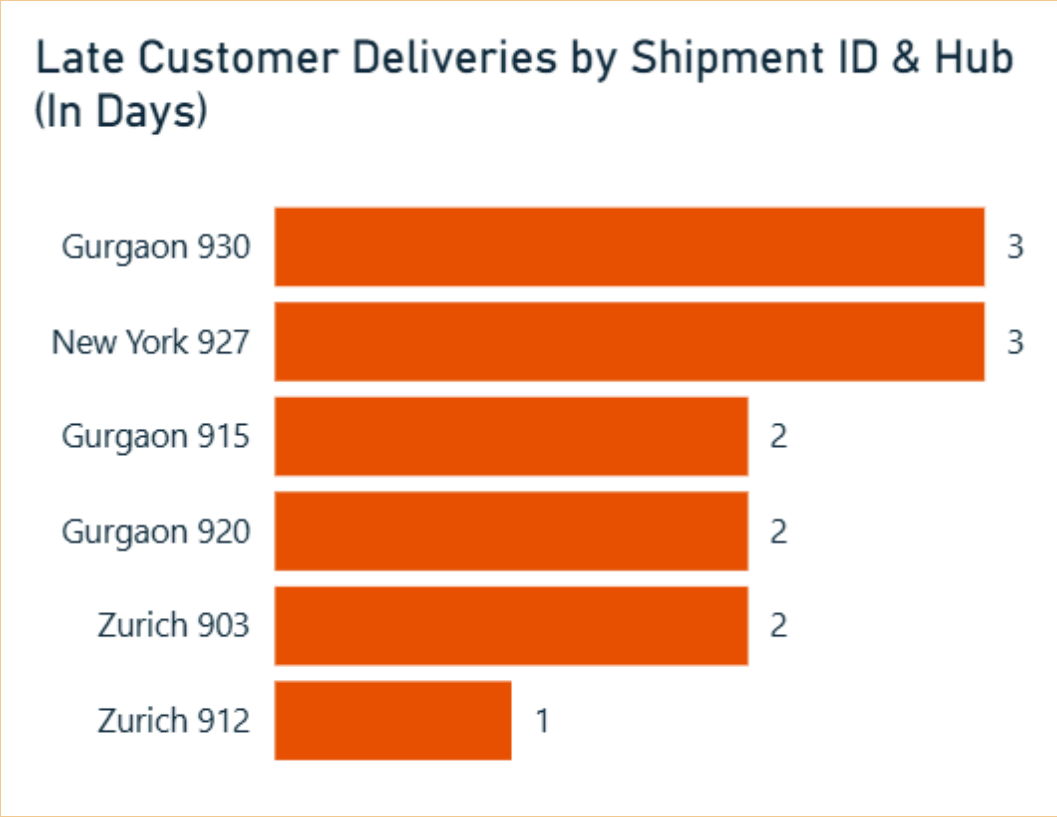
Cold-Chain SKUs by Unit Cost (INR)



Q2) CUSTOMER SHIPMENT DELAY ANALYSIS

“Our logistics operations team needs a report on customer orders that suffered delivery delays in 2025. Please write a query to list all customer shipments that were delivered late. We need to see the shipment details, the warehouse location, the promised delivery date, the actual delivery date, and how many days late the delivery was. Order the results starting from the most delayed shipment down to the least delayed.”

```
SELECT
S.shipment_id,
S.region,
W.city AS Warehouse_location,
S.promised_delivery_date,
S.actual_delivery_date,
DATEDIFF(S.actual_delivery_date,S.promised_delivery_date) AS Delay_in_days
FROM fact_customer_shipments AS S
JOIN dim_warehouses AS W
ON S.warehouse_id = W.warehouse_id
WHERE S.actual_delivery_date > S.promised_delivery_date
ORDER BY Delay_in_days DESC;
```



Result

Identified late customer shipments with maximum delays reaching up to 3 days across APAC, EU, and NA fulfillment hubs.

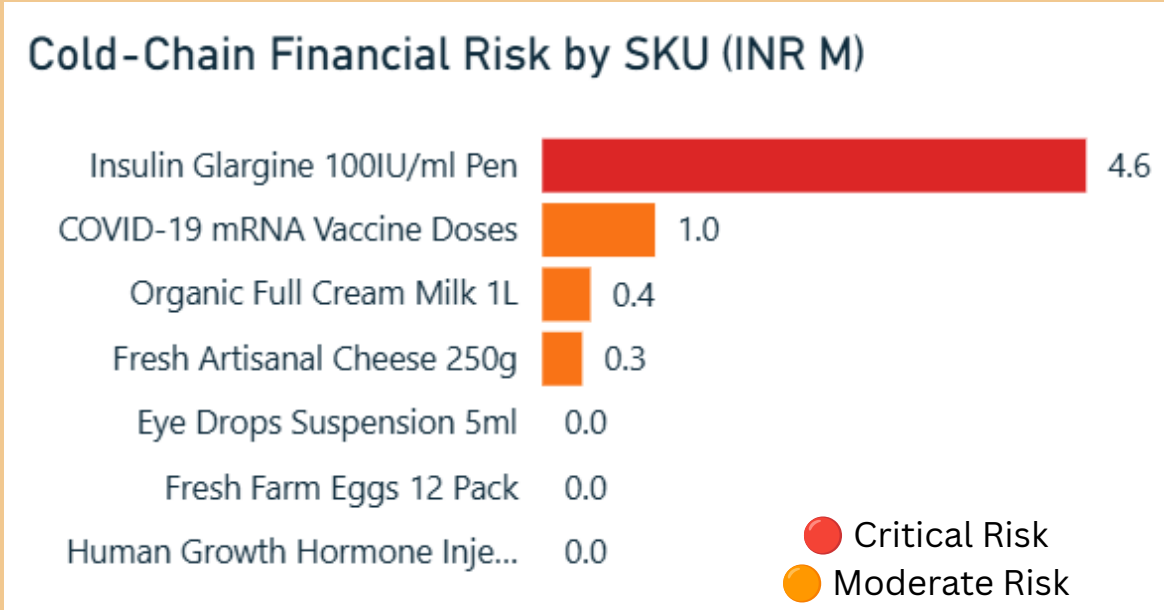
Q3) COLD-CHAIN TEMPERATURE BREACH & RISK AUDIT

"Our cold-chain logistics team requires an urgent audit of all temperature-controlled shipments. Please generate a report identifying all products where recorded transit temperatures breached safe target storage bounds (either below minimum or above maximum limits).For each cold-chain SKU, calculate total shipments count, total breached units, and total financial inventory value at risk in Millions (INR M), sorted by financial risk descending."

```
SELECT
  p.product_name,
  p.category,
  COUNT(s.shipment_id) AS Total_shipment_count,
  SUM(CASE WHEN S.avg_transit_temp_c < P.target_temp_min_c OR s.avg_transit_temp_c > p.target_temp_max_c
    THEN 1 ELSE 0 END)
  AS Total_Breached_Shipments,
  SUM(CASE WHEN s.avg_transit_temp_c < p.target_temp_min_c OR s.avg_transit_temp_c > p.target_temp_max_c
    THEN s.units_shipped ELSE 0 END)
  AS Total_Breached_Units,
  ROUND(SUM(CASE WHEN s.avg_transit_temp_c < p.target_temp_min_c OR s.avg_transit_temp_c > p. target_temp_max_c
    THEN s.units_shipped*p.unit_cost ELSE 0 END)/1000000 ,2 )
  AS Financial_Risk_Val_Millions
FROM dim_products AS p
JOIN fact_customer_shipments AS s
ON p.product_id=s.product_id
WHERE p.is_cold_chain = 1
GROUP BY p.product_name,p.category
ORDER BY Financial_Risk_Val_Millions;
```

Result

Identified a total inventory risk of ₹6.34 Million (INR M) across 4 breached cold-chain categories. Crucially, uncovered that Insulin Glargine alone accounted for 73% of the total financial exposure (₹4.63M), prioritizing IoT temperature sensors on high-value pharma delivery routes."



Q4. SUPPLIER OTIF (ON-TIME IN-FULL) PERFORMANCE AUDIT

"Our global procurement team requires an evaluation of vendor delivery performance across all 2025 purchase orders. Please generate a report measuring each supplier's On-Time In-Full (OTIF) success rate, where an order is considered compliant only if delivered on or before lead-time date AND in full quantity. Display supplier details, total orders placed, total OTIF orders delivered, OTIF success percentage, and operational status flagging vendors below 80% as 'CONTRACT_REVIEW_REQUIRED', sorted by OTIF percentage descending."

```
SELECT
CONCAT_WS('|',s.supplier_name,s.country) AS Supplier_detail,
COUNT(o.po_id) AS Total_Purchase_Orders_Placed,
SUM( CASE WHEN datediff(o.delivery_date,order_date)<=S.lead_time_days
OR o.received_qty >= o.order_qty THEN 1 ELSE 0 END )AS Total_OTIF_Orders,

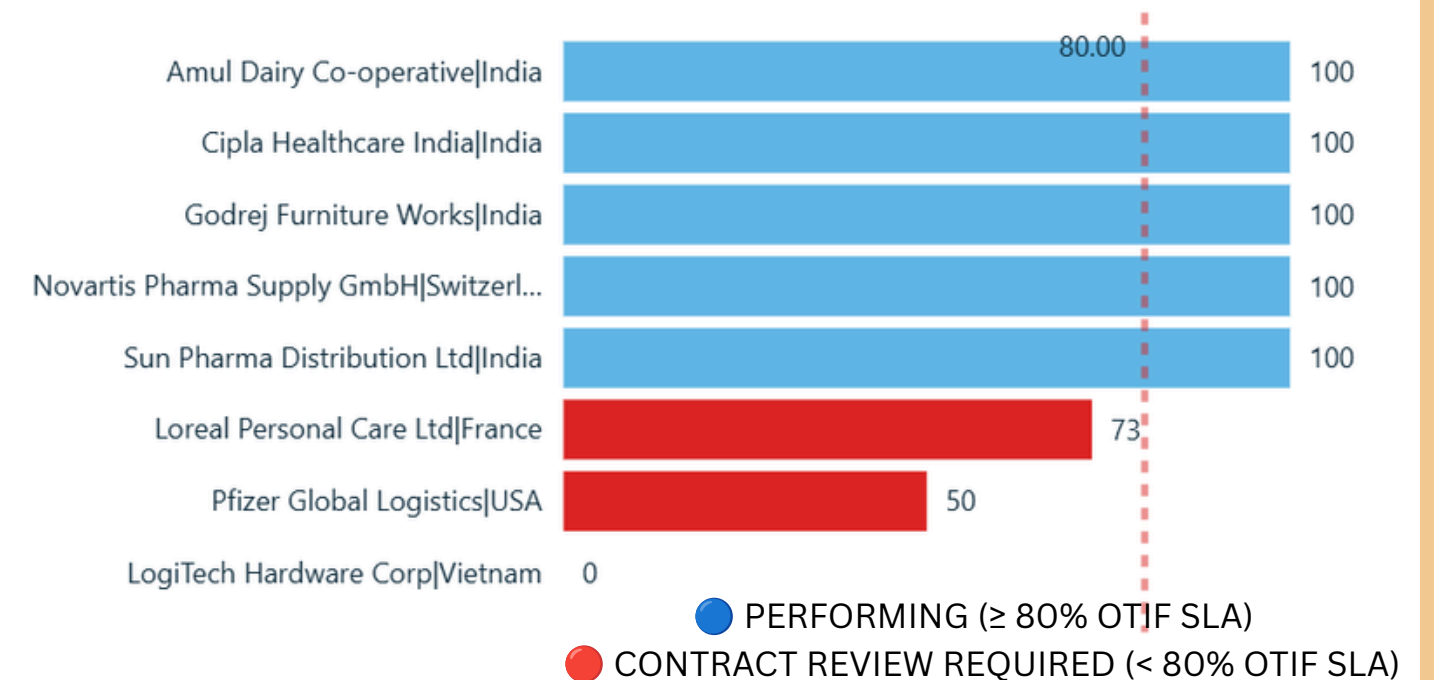
ROUND(SUM( CASE WHEN datediff(o.delivery_date,order_date)<=S.lead_time_days
OR o.received_qty >= o.order_qty THEN 1 ELSE 0 END)*100/ COUNT(o.po_id),2) AS OTIF_PCT,

CASE WHEN(
SUM( CASE WHEN datediff(o.delivery_date,order_date)<=S.lead_time_days
OR o.received_qty >= o.order_qty THEN 1 ELSE 0 END)*100/ COUNT(o.po_id)) < 80
THEN 'CONTRACT_REVIEW_REQUIRED'ELSE 'PERFORMING' END AS SUPPLIER_FLAG
FROM dim_suppliers AS s
JOIN fact_purchase_orders AS o ON S.supplier_id= O.supplier_id
WHERE o.po_status ='delivered'
GROUP BY s.supplier_name,s.supplier_name,s.country
ORDER BY OTIF_PCT DESC;
```

Result

Evaluated all global suppliers across 100 purchase orders and successfully isolated 3 underperforming vendors failing the 80% OTIF SLA benchmark (LogiTech 0%, Pfizer 50%, Loreal 72.73%), enabling procurement leadership to initiate vendor contract renegotiations.

Supplier OTIF SLA Success Rate (%)



Q5) TOP PRODUCT VOLUME RANKING PER CATEGORY

"Our product category managers need to identify the top-performing SKUs within each product category based on total units shipped in 2025. Please generate a report that identifies and ranks the Top 2 highest-volume products for every category, displaying the category name, product name, total units shipped, and category rank order, sorted by category and rank."

```
WITH CTE1 AS(  
  SELECT  
    P.product_name,  
    p.category,  
    SUM(s.units_shipped) as Total_unit_shipped,  
    DENSE_RANK() OVER( partition by p.CATEGORY order by SUM(s.units_shipped) desc) as Top_Sku  
  FROM dim_products AS p  
  JOIN fact_customer_shipments AS s  
  ON p.product_id=s.product_id  
  GROUP BY p.product_name,p.category)  
  
SELECT * FROM CTE1  
WHERE TOP_SKU <=2  
ORDER BY category ASC ,Top_Sku ASC;
```

product_name	category	Total_unit_shipped	Top_Sku
Wireless Noise Canceling Mic	Electronics	800	1
AtIQ Laptop Pro 15 Inch	Electronics	350	2
Organic Full Cream Milk 1L	FMCG Fresh	15000	1
Fresh Artisanal Cheese 250g	FMCG Fresh	2200	2
Ergonomic Office Task Chair	Furniture	100	1
Hydrating Face Serum 30ml	Personal Care	1500	1
SPF 50 Sunscreen Lotion 100ml	Personal Care	1200	2
Insulin Glargine 100IU/ml Pen	Pharma Cold-Chain	7800	1
COVID-19 mRNA Vaccine Doses	Pharma Cold-Chain	1500	2
Paracetamol 650mg 150 Tabs Pack	Pharma General	11000	1
Amoxicillin Antibiotic 500mg	Pharma General	7000	2



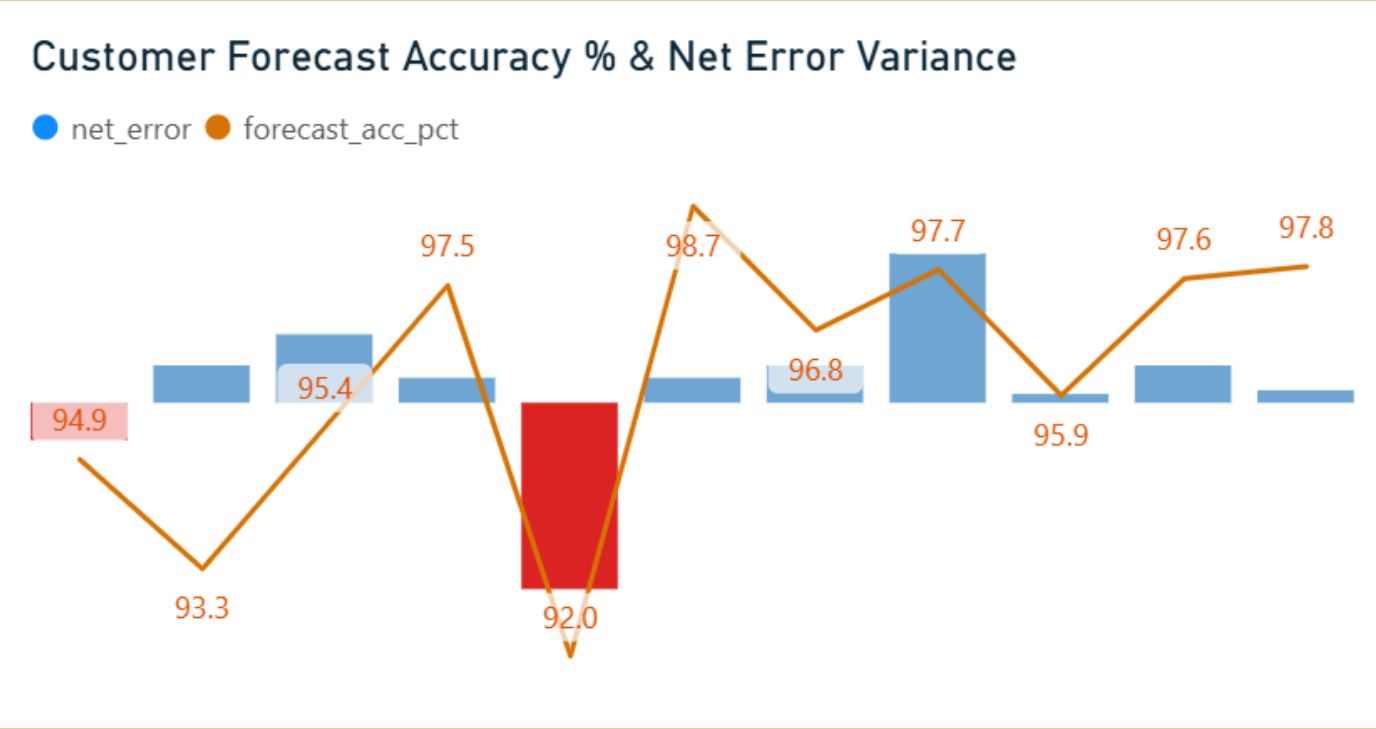
Result

Engineered a Window Function (DENSE_RANK OVER PARTITION) within a CTE to isolate the Top 2 volume drivers per category (e.g., Insulin Glargine & mRNA Vaccines in Pharma Cold-Chain; Milk & Cheese in FMCG Fresh), providing inventory leads with targeted restocking priorities and mitigating overstocking risks.

Q6) CUSTOMER FORECAST ACCURACY & MAPE ANALYSIS

"Our demand planning team requires an executive forecast accuracy report across all customer accounts for 2025. Please generate a report evaluating total forecasted quantity against total actual sold quantity per customer account. Calculate net forecast error, absolute forecast error (MAPE), and final forecast accuracy percentage (ensuring severe error rates above 100% are capped appropriately), sorted by forecast accuracy percentage descending."

```
WITH CTE1 AS
(
  SELECT
    customer_code,
    product_id,
    SUM(forecast_quantity) AS Total_forecast_quantity,
    SUM(sold_quantity) AS Total_sold_quantity,
    SUM(forecast_quantity-sold_quantity) AS net_error,
    SUM(ABS(forecast_quantity -sold_quantity)) AS abs_error,
    SUM(ABS(forecast_quantity -sold_quantity))*100/SUM(forecast_quantity) AS abs_error_pct
  FROM fact_forecast_monthly
  GROUP BY customer_code,product_id)
SELECT
  *, (100-abs_error_pct) AS forecast_acc_pct
FROM CTE1
```



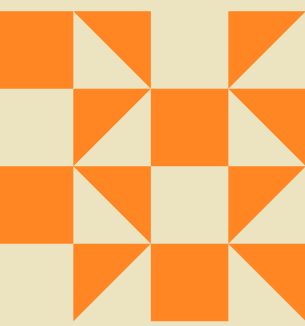
Result

Delivered an executive forecast accuracy report across all customer accounts. Crucially uncovered that while Net Error appeared low due to error cancellation, Absolute Error (MAPE) revealed true operational variances of 900 units on Customer 70002017 and 3,500 units on Customer 70002019, allowing demand planners to recalibrate baseline forecasting algorithms.



STRATEGIC RECOMMENDATIONS & EXECUTIVE SUMMARY

1. ❄️ Cold-Chain IoT Protection: Deploy real-time IoT sensors on high-value Pharma delivery routes to protect Insulin Glargine (saving ₹4.63M in potential write-offs).
2. 📦 Vendor SLA Enforcement: Initiate contract review for the 3 underperforming suppliers (LogiTech 0%, Pfizer 50%, Loreal 72.73%) failing the 80% OTIF SLA benchmark.
3. 📊 Demand Forecast Recalibration: Mandate Absolute Error (MAPE) metrics across demand planning teams to eliminate error cancellation and optimize safety stock



Thank You!
For your time & attention

Let's Connect

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